Chapter 3. The Multivariate Normal Distribution

Definition and properties, conditional distributions, best linear prediction, multiple and partial correlation, quadratic forms

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1 Definition

1.1 Non-singular case

The univariate normal density with mean μ and variance σ^2 is $f(x) = (2\pi\sigma^2)^{-1/2}\exp[-\frac{1}{2}(x-\mu)\sigma^{-2}(x-\mu)]$. About 68.3% of its mass lies within one standard deviation of μ and 95.4% within two. Replacing $(x-\mu)^2/\sigma^2$ by a squared Mahalanobis distance gives the multivariate version.

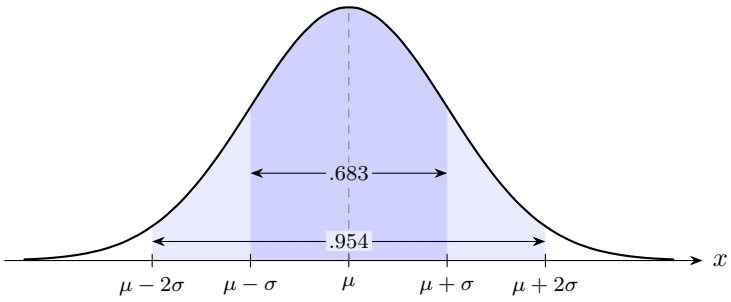


Figure. *The $N(\mu,\sigma^2)$ density. The dark band $\mu\pm\sigma$ carries probability* 0.683 *and the light band $\mu\pm 2\sigma$ carries* 0.954.

Definition 3.1: multivariate normal (non-singular)

For $\boldsymbol{\mu}\in\mathbb{R}^p$ and $\boldsymbol{\Sigma}_{(p\times p)}\succ 0$, $\mathbf{X}\sim N_p(\boldsymbol{\mu},\boldsymbol{\Sigma})$ means $\mathbf{X}$ has density

$$f(\mathbf{x}) = |2\pi\boldsymbol{\Sigma}|^{-1/2}\exp\Big[-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big], \qquad \mathbf{x}\in\mathbb{R}^p,$$

where $|2\pi\boldsymbol{\Sigma}| = (2\pi)^p|\boldsymbol{\Sigma}|$.

One checks that f is a valid joint density on $\mathbb{R}^p$: $f(\mathbf{x})>0$ for all $\mathbf{x}$, and $\int f(\mathbf{x})\,d\mathbf{x}=1$.

Proof that $\int f = 1$.

Step 1 (centering and diagonalizing). With $\boldsymbol{\Sigma} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$, $\mathbf{Q}$ orthogonal and $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1,\dots,\lambda_p)$, put $\mathbf{z} = \mathbf{Q}^\top(\mathbf{x}-\boldsymbol{\mu})$. Then $(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = \mathbf{z}^\top\boldsymbol{\Lambda}^{-1}\mathbf{z} = \sum_i z_i^2/\lambda_i$, $|\boldsymbol{\Sigma}| = \prod_i\lambda_i$, and the Jacobian is $|\mathbf{Q}| = \pm 1$.

Step 2 (product of univariate integrals). The exponent separates, so

$$\int_{\mathbb{R}^p} f(\mathbf{x})\,d\mathbf{x} = \prod_{i=1}^p \int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\lambda_i}}\exp\Big(-\frac{z_i^2}{2\lambda_i}\Big)dz_i = \prod_{i=1}^p 1 = 1.$$

Equivalently, $\mathbf{Y} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{X}-\boldsymbol{\mu})$ has $d\mathbf{y}/d\mathbf{x} = |\boldsymbol{\Sigma}|^{-1/2}$, and

$$\int f\,d\mathbf{x} = \int\cdots\int (2\pi)^{-p/2}|\boldsymbol{\Sigma}|^{-1/2}e^{-\mathbf{y}^\top\mathbf{y}/2}|\boldsymbol{\Sigma}|^{1/2}d\mathbf{y} = \prod_{i=1}^p\int(2\pi)^{-1/2}e^{-y_i^2/2}dy_i = 1.$$

■

The substitution shows $\mathbf{Y} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{X}-\boldsymbol{\mu})\sim N_p(\mathbf{0},\mathbf{I}_p)$, i.e. $\mathbf{X}\overset{d}{=}\boldsymbol{\Sigma}^{1/2}\mathbf{Y}+\boldsymbol{\mu}$. Hence $\mathrm{E}\mathbf{X}=\boldsymbol{\mu}$, $\mathrm{Var}\mathbf{X}=\boldsymbol{\Sigma}$, and

$$U = (\mathbf{X}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) = \mathbf{Y}^\top\mathbf{Y} = \textstyle\sum_i Y_i^2\sim\chi^2(p).$$

1.2 General definition and Cramér–Wold

The characteristic function of a random vector $\mathbf{X}$ is $\varphi_{\mathbf{X}}(\mathbf{t}) = \mathrm{E}\exp(i\mathbf{t}^\top\mathbf{X})$, $\mathbf{t}\in\mathbb{R}^p$. It is $\mathbb{C}$-valued and always exists, and (Theorem 2.25): (1) $\varphi_{\mathbf{X}}=\varphi_{\mathbf{Y}}$ iff $\mathbf{X}\overset{d}{=}\mathbf{Y}$; (2) $\mathbf{X}_n\Rightarrow\mathbf{X}$ iff $\varphi_{\mathbf{X}_n}(\mathbf{t})\to\varphi_{\mathbf{X}}(\mathbf{t})$

for all $\mathbf{t}$.

Remark 3.2: Cramér–Wold device (Corollary 2)

If $\mathbf{X}$ is a $p \times 1$ random vector, its distribution is uniquely determined by the distributions of the linear functions $\mathbf{t}^\top \mathbf{X}$, $\mathbf{t} \in \mathbb{R}^p$: the univariate laws of all linear combinations determine the multivariate joint law.

Proof. Let $F_{\mathbf{t}}$ be the distribution function of $\mathbf{t}^\top \mathbf{X}$. Then $\varphi_{\mathbf{X}}(\mathbf{t}) = \mathrm{E}[e^{i\mathbf{t}^\top \mathbf{X}}] = \int_{\mathbb{R}} e^{iu}\, dF_{\mathbf{t}}(u)$, so knowing $F_{\mathbf{t}}$ for every $\mathbf{t}$ determines $\varphi_{\mathbf{X}}$, which by (1) determines the distribution of $\mathbf{X}$. ■

Definition 3.3: multivariate normal (singular allowed)

$\mathbf{X} \in \mathbb{R}^p$ has a multivariate normal distribution iff $\mathbf{t}^\top \mathbf{X}$ is univariate normal for every $\mathbf{t} \in \mathbb{R}^p$ (including the point mass $N(c, 0)$).

This definition does not require $\mathrm{Cov}(\mathbf{X})$ to be non-singular: $\mathbf{X}$ is MVN iff *every* projection onto a one-dimensional subspace is normal. It is well defined because, by the Cramér–Wold device, the distributions of all $\mathbf{t}^\top \mathbf{X}$ determine that of $\mathbf{X}$.

Example 3.4: normal marginals do not imply joint normality (Example 4.8)

$X_1 \sim N(0, 1)$ and $X_2 = -X_1$ if $-1 \le X_1 \le 1$, $X_2 = X_1$ otherwise.
(a) $X_2 \sim N(0, 1)$.
(b) (X_1, X_2) is not bivariate normal.

Proof. (a) X_1 is standard normal and symmetric about zero, so the map $X_1 \mapsto -X_1$ on $\{-1 \le X_1 \le 1\}$ does not change the distribution. For $-1 \le X_1 \le 1$, $P[X_2 \le x_2 \mid -1 \le X_1 \le 1] = P[-X_1 \le x_2 \mid -1 \le X_1 \le 1] = P[X_1 \le x_2 \mid -1 \le X_1 \le 1]$ by symmetry, and for $|X_1| > 1$, $P[X_2 \le x_2 \mid |X_1| > 1] = P[X_1 \le x_2 \mid |X_1| > 1]$. Combining, $P[X_2 \le x_2] = P[X_1 \le x_2]$, so $X_2 \sim N(0, 1)$.

(b) Consider the linear combination $X_1 - X_2$. For $-1 \le X_1 \le 1$ it equals $2X_1$, and otherwise $X_1 - X_2 = 0$. Since a linear combination of X_1 and X_2 is not normal, (X_1, X_2) cannot be jointly normal. (Concretely, $P(X_1 - X_2 = 0) = P(|X_1| > 1) \approx 0.317 \in (0, 1)$, so $X_1 - X_2$ is neither a continuous normal nor a point mass.) ■

1.3 Characteristic function

Theorem 3.5: characteristic function of the MVN

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succeq 0$, then $\varphi_{\mathbf{X}}(\mathbf{t}) = \exp\left(i\mathbf{t}^\top \boldsymbol{\mu} - \tfrac{1}{2}\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}\right)$. If $\boldsymbol{\Sigma} \succ 0$, the density above exists.

Proof. $\mathbf{t}^\top \mathbf{X} \sim N(\mathbf{t}^\top \boldsymbol{\mu}, \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$ and $\varphi_{N(m,v)}(s) = \exp(ism - \tfrac{1}{2}s^2 v)$, so $\varphi_{\mathbf{X}}(\mathbf{t}) = \varphi_{\mathbf{t}^\top \mathbf{X}}(1) = \exp(i\mathbf{t}^\top \boldsymbol{\mu} - \tfrac{1}{2}\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$. ■

[Supplement] Since the characteristic function depends only on $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the notation $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is justified under the general definition as well.

2 Geometry: Constant-Density Ellipsoids

Location parameter $\boldsymbol{\mu}$, shape parameter $\boldsymbol{\Sigma} \succ 0$. The contours of constant density are

$$E_c = \{\mathbf{x} \in \mathbb{R}^p : f(\mathbf{x}) = c_0\} = \{\mathbf{x} : (\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = c^2\},$$

where c is the Mahalanobis distance $d_M(\boldsymbol{\mu}, \mathbf{x})$. With $\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\lambda_1 \geq \cdots \geq \lambda_p > 0$, E_c is an ellipsoid centered at $\boldsymbol{\mu}$ with principal axes along $\mathbf{u}_i$ and half-lengths proportional to $\sqrt{\lambda_i}$. If $\boldsymbol{\Sigma} = \mathbf{I}_p$ it is a sphere of radius c.

Example 3.6: bivariate normal

$N_2(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \mathbf{R}_{\pi/4}\mathrm{diag}(3,1)\mathbf{R}_{\pi/4}^\top$, $\mathbf{R}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$. Location: the origin $\boldsymbol{\mu} = \mathbf{0}$. Shape: elliptical contours with principal axes at $45°$ and $-45°$ and half-axis ratio $\sqrt{3} : 1$.

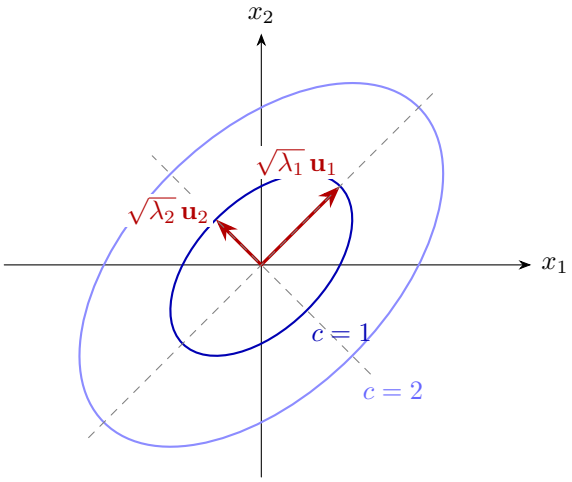


Figure. *Contours* $(\mathbf{x}-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) = c^2$ *of the bivariate normal in the example* ($\lambda_1 = 3$, $\lambda_2 = 1$). *The half-axes are* $c\sqrt{\lambda_i}$ *along the eigenvectors* $\mathbf{u}_1 = (1,1)^\top/\sqrt{2}$ *and* $\mathbf{u}_2 = (-1,1)^\top/\sqrt{2}$. *In higher dimensions ellipsoids play the same role.*

[Supplement] The sign and size of the correlation control the tilt and the eccentricity of the ellipses. For $\boldsymbol{\Sigma} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ the eigenvalues are $1 \pm \rho$ with eigenvectors $(1, \pm 1)^\top/\sqrt{2}$, so the contours are circles when $\rho = 0$ and become thinner along the $45°$ (or $-45°$) line as $|\rho| \to 1$.

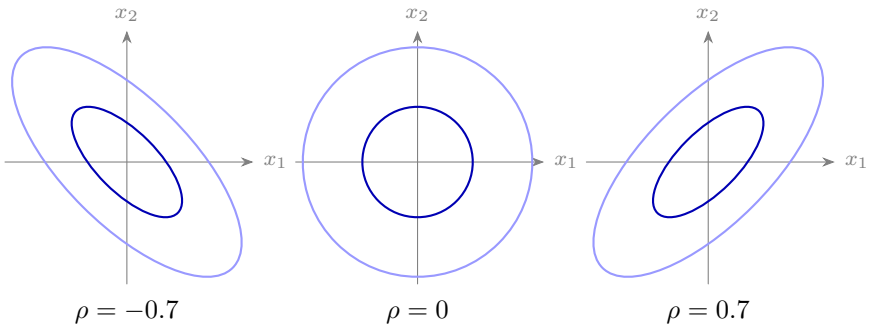


Figure. *[Supplement] Contours containing probability* 0.5 *(dark) and* 0.95 *(light) for unit variances and* $\rho = -0.7, 0, 0.7$. *Positive correlation stretches the ellipse along* $x_2 = x_1$, *negative correlation along* $x_2 = -x_1$.

3 Linear Transformations, Subsets, Independence, Conditionals

Motivation. The multivariate normal is the default model because it is closed under every operation we perform: linear maps, marginals and conditionals are again normal, and zero correlation means independence. The conditional distribution formula is the population version of linear regression.

Theorem 3.7: linear combinations (Result 4.2)

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then $\mathbf{a}^\top \mathbf{X} = a_1 X_1 + \cdots + a_p X_p \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$. Conversely, if $\mathbf{a}^\top \mathbf{X} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$ for every $\mathbf{a}$, then $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Proof. $\mathrm{E}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\mu}$ and $\mathrm{Var}(\mathbf{a}^\top \mathbf{X}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$. Normality via the MGF: $M_{\mathbf{a}^\top \mathbf{X}}(t) = \mathrm{E} e^{t \mathbf{a}^\top \mathbf{X}} = M_{\mathbf{X}}(t\mathbf{a}) = \exp(t \mathbf{a}^\top \boldsymbol{\mu} + \frac{1}{2} t^2 \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$, the MGF of $N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$. The converse is the general definition. ■

Theorem 3.8: linear transformations (Result 4.3)

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\mathbf{A} \in \mathbb{R}^{q \times p}$ and $\mathbf{d} \in \mathbb{R}^p$, then $\mathbf{A}\mathbf{X} \sim N_q(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top)$ and $\mathbf{X} + \mathbf{d} \sim N_p(\boldsymbol{\mu} + \mathbf{d}, \boldsymbol{\Sigma})$.

Proof. $\mathrm{E}(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$. Any $\mathbf{b}^\top(\mathbf{A}\mathbf{X}) = (\mathbf{A}^\top \mathbf{b})^\top \mathbf{X}$ is a linear combination of $\mathbf{X}$, normal by Theorem 3.7, so $\mathbf{A}\mathbf{X}$ is MVN. $\mathbf{a}^\top(\mathbf{X} + \mathbf{d}) = \mathbf{a}^\top \mathbf{X} + \mathbf{a}^\top \mathbf{d}$ only shifts the mean. ■

Theorem 3.9: subsets (Result 4.4)

Partition $\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix}$, $\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}$, $\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}$. Then $\mathbf{X}_1 \sim N_q(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$; all subsets of $\mathbf{X}$ are normal.

Proof. Theorem 3.8 with $\mathbf{A} = [\mathbf{I}_{q \times q} \ \mathbf{0}_{q \times (p-q)}]$. Any subset can be relabelled as $\mathbf{X}_1$. ■

Theorem 3.10: independence and covariance (Result 4.5)

(a) If $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2$, then $\mathrm{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \mathbf{0}$.

(b) If $\begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix} \sim N_{q_1+q_2}$, then $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2 \iff \boldsymbol{\Sigma}_{12} = \mathbf{0}$.

(c) If $\mathbf{X}_1 \sim N_{q_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$ and $\mathbf{X}_2 \sim N_{q_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$ are independent, then $(\mathbf{X}_1^\top, \mathbf{X}_2^\top)^\top \sim N_{q_1+q_2}\Big(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{22} \end{pmatrix} \Big)$.

Proof.

(a) [Supplement] $\mathrm{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \mathrm{E}[(\mathbf{X}_1 - \boldsymbol{\mu}_1)(\mathbf{X}_2 - \boldsymbol{\mu}_2)^\top]$, and by independence the expectation of each entry factorizes: $\mathrm{E}[(X_{1i} - \mu_{1i})(X_{2j} - \mu_{2j})] = \mathrm{E}(X_{1i} - \mu_{1i})\,\mathrm{E}(X_{2j} - \mu_{2j}) = 0$.

(b) The joint characteristic function is

$$\varphi_{\mathbf{X}}(\mathbf{t}_1, \mathbf{t}_2) = \exp\Big\{ i(\mathbf{t}_1^\top \boldsymbol{\mu}_1 + \mathbf{t}_2^\top \boldsymbol{\mu}_2) - \tfrac{1}{2} \begin{pmatrix} \mathbf{t}_1 \\ \mathbf{t}_2 \end{pmatrix}^\top \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{t}_1 \\ \mathbf{t}_2 \end{pmatrix} \Big\}.$$

If $\boldsymbol{\Sigma}_{12} = \mathbf{0}$ it factorizes as $\varphi_{\mathbf{X}_1}(\mathbf{t}_1)\varphi_{\mathbf{X}_2}(\mathbf{t}_2)$: independence. Conversely, independence forces the factorization, so the cross term $\mathbf{t}_1^\top \boldsymbol{\Sigma}_{12} \mathbf{t}_2$ must vanish for all $\mathbf{t}$, i.e. $\boldsymbol{\Sigma}_{12} = \mathbf{0}$.

(c) [Supplement] Independence gives $\varphi_{\mathbf{X}}(\mathbf{t}_1,\mathbf{t}_2) = \varphi_{\mathbf{X}_1}(\mathbf{t}_1)\varphi_{\mathbf{X}_2}(\mathbf{t}_2)$, which is the characteristic function above with $\boldsymbol{\Sigma}_{12} = \mathbf{0}$, i.e. that of the stated normal law. ■

Remark 3.11: "uncorrelated ⇒ independent" needs joint normality

[Supplement] In Example 3.4 each marginal is normal but the pair is not jointly normal, so (b) does not apply.

Theorem 3.12: conditional distribution (Result 4.6)

If $\boldsymbol{\Sigma}_{22} \succ 0$,

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \;\sim\; N_r\big(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2),\; \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\big).$$

The conditional mean is linear in $\mathbf{x}_2$; the conditional covariance $\boldsymbol{\Sigma}_{11\cdot 2} := \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ does not depend on $\mathbf{x}_2$.

Proof. Define $\mathbf{X}_1^* = \mathbf{X}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$, $\mathbf{X}_2^* = \mathbf{X}_2$, i.e. $\mathbf{X}^* = \mathbf{A}\mathbf{X}$ with $\mathbf{A} = \begin{pmatrix} \mathbf{I}_r & -\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1} \\ \mathbf{0} & \mathbf{I}_s \end{pmatrix}$. By Theorem 3.8 $\mathbf{X}^*$ is MVN and

$$\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top = \begin{pmatrix} \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{22} \end{pmatrix}.$$

$\mathrm{Cov}(\mathbf{X}_1^*, \mathbf{X}_2^*) = \mathbf{0}$, so $\mathbf{X}_1^* \perp\!\!\!\perp \mathbf{X}_2$ (Theorem 3.10(b)) and $\mathbf{X}_1^* \sim N_r(\boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{11\cdot 2})$. Since $\mathbf{X}_1 = \mathbf{X}_1^* + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$, conditioning on $\mathbf{X}_2 = \mathbf{x}_2$ makes the second term constant without changing the law of $\mathbf{X}_1^*$: $\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \overset{d}{=} \mathbf{X}_1^* + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{x}_2$. Therefore $\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \sim N_r(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2),\; \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21})$. ■

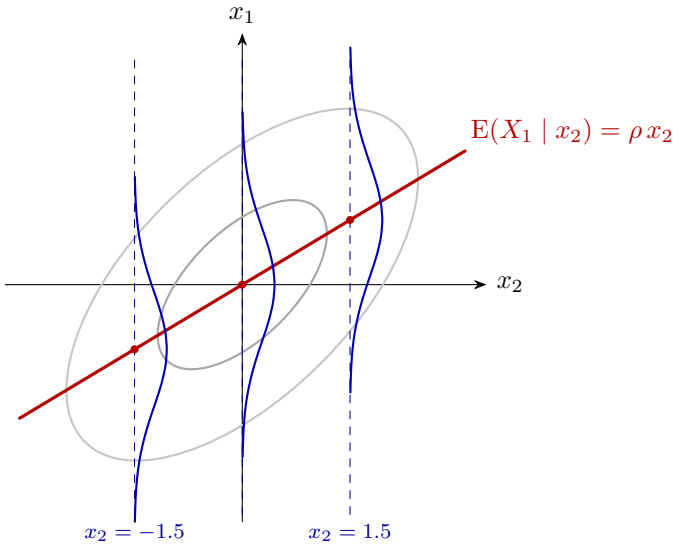


Figure. *Result 4.6 for a standardized pair with $\rho = 0.6$. Each vertical slice $X_2 = x_2$ of the joint density (gray contours) is a normal density (blue) centred on the regression line $\mu_1 + \sigma_{12}\sigma_{22}^{-1}(x_2 - \mu_2) = 0.6\,x_2$, with the same variance $\sigma_{11\cdot 2} = 1 - \rho^2 = 0.64$ for every x_2.*

Remark 3.13: residuals, projections and partial variance

$\mathbf{X}_1^* = \mathbf{X}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\mathbf{X}_2$ is (up to centring) the residual $\mathbf{X}_1 - \mathrm{BLP}(\mathbf{X}_1 \mid \mathbf{X}_2)$ after projecting $\mathbf{X}_1$ onto the span of $\mathbf{X}_2$. The residual is uncorrelated with $\mathbf{X}_2$, and independent of it in the Gaussian case. $\sigma^2_{11\cdot 2} = \sigma_{11} - \boldsymbol{\sigma}_{21}^\top \boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\sigma}_{21}$ (more generally $\boldsymbol{\Sigma}_{11\cdot 2}$) is the **partial variance** of $\mathbf{X}_1$ after removing the effect of $\mathbf{X}_2$, i.e. $\mathrm{Var}(\mathbf{X}_1^*)$.

4 Best Linear Prediction, Multiple and Partial Correlation

Motivation. How well can Y be predicted linearly from $\mathbf{X}$, and how correlated are two variables once a third is held fixed? Both answers come from the conditional covariance of the normal distribution.

Let $\mathbf{X} \in \mathbb{R}^q$ be the predictor and $Y \in \mathbb{R}$ the response, $\mathbf{U} = (\mathbf{X}^\top, Y)^\top \in \mathbb{R}^p$ $(p = q+1)$,

$$\boldsymbol{\mu} = \mathrm{E}(\mathbf{U}) = \begin{pmatrix} \boldsymbol{\mu}_X \\ \mu_Y \end{pmatrix}, \qquad \boldsymbol{\Sigma} = \mathrm{Var}(\mathbf{U}) = \begin{pmatrix} \boldsymbol{\Sigma}_{XX} & \boldsymbol{\Sigma}_{XY} \\ \boldsymbol{\Sigma}_{YX} & \sigma_Y^2 \end{pmatrix},$$

with sample counterparts $\bar{\mathbf{u}} = (\bar{\mathbf{x}}^\top, \bar{y})^\top$ and $\mathbf{S} = \begin{pmatrix} \mathbf{S}_{XX} & \mathbf{S}_{XY} \\ \mathbf{S}_{YX} & s_Y^2 \end{pmatrix}$.

4.1 Best linear predictor

Theorem 3.14: best linear predictor

$\mathrm{BLP}(Y \mid \mathbf{X}) := \mathbf{A}^*\mathbf{X} + b^*$, $(\mathbf{A}^*, b^*) = \arg\min_{\mathbf{A}\in\mathbb{R}^{1\times q},\, b\in\mathbb{R}} \mathrm{E}\{Y - (\mathbf{A}\mathbf{X} + b)\}^2$. With finite second moments and $\boldsymbol{\Sigma}_{XX} \succ 0$,

$$\mathbf{A}^* = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}, \qquad b^* = \mu_Y - \mathbf{A}^*\boldsymbol{\mu}_X, \qquad \mathrm{BLP}(Y \mid \mathbf{X}) = \mu_Y + \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}(\mathbf{X} - \boldsymbol{\mu}_X).$$

Proof. Let $W := Y - \mu_Y - \mathbf{A}^*(\mathbf{X} - \boldsymbol{\mu}_X)$; then $\mathrm{E}W = 0$ and $\mathrm{Cov}(W, \mathbf{X}) = \boldsymbol{\Sigma}_{YX} - \mathbf{A}^*\boldsymbol{\Sigma}_{XX} = \mathbf{0}$. For any $(\mathbf{A}, b)$,

$$Y - \mathbf{A}\mathbf{X} - b = W + (\mathbf{A}^* - \mathbf{A})(\mathbf{X} - \boldsymbol{\mu}_X) + \{\mu_Y - \mathbf{A}\boldsymbol{\mu}_X - b\},$$

three mutually uncorrelated terms (the last is constant), so

$$\mathrm{E}(Y - \mathbf{A}\mathbf{X} - b)^2 = \mathrm{E}W^2 + (\mathbf{A}^* - \mathbf{A})\boldsymbol{\Sigma}_{XX}(\mathbf{A}^* - \mathbf{A})^\top + (\mu_Y - \mathbf{A}\boldsymbol{\mu}_X - b)^2 \geq \mathrm{E}W^2,$$

with equality iff $\mathbf{A} = \mathbf{A}^*$ and $b = \mu_Y - \mathbf{A}^*\boldsymbol{\mu}_X$. ■

Remark 3.15: properties of the BLP

$\mathrm{E}[\mathrm{BLP}(Y \mid \mathbf{X})] = \mu_Y$, $\mathrm{Var}[\mathrm{BLP}(Y \mid \mathbf{X})] = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$; the residual $W = Y - \mathrm{BLP}(Y \mid \mathbf{X})$ has $\mathrm{E}W = 0$, $\mathrm{Var}W = \sigma_Y^2 - \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ and $\mathrm{Cov}(W, \mathrm{BLP}) = 0$. Under joint normality $\mathrm{BLP}(Y \mid \mathbf{X}) = \mathrm{E}(Y \mid \mathbf{X})$ (Theorem 3.12). In block notation $\mathrm{E}[\mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_1)] = \boldsymbol{\mu}_2$, $\mathrm{Var}[\mathrm{BLP}(\mathbf{X}_2 \mid \mathbf{X}_1)] = \boldsymbol{\Sigma}_{21}\boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12}$.

4.2 Multiple correlation

Definition 3.16: multiple correlation of Y with $\mathbf{X}$

$$\rho_{\text{mult}} = \text{mcorr}(\mathbf{X}, Y) := \max_{\mathbf{a}\in\mathbb{R}^q\setminus\{\mathbf{0}\}} \text{Corr}(Y, \mathbf{a}^\top\mathbf{X}), \qquad \rho^2_{\text{mult}} = \frac{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}}{\sigma_Y^2}.$$

With $\hat{Y} = \text{BLP}(Y \mid \mathbf{X})$, $\rho_{\text{mult}} = \text{Corr}(Y, \hat{Y})$.

Proof. $\text{Corr}(Y, \mathbf{a}^\top\mathbf{X} + b) = \dfrac{\mathbf{a}^\top\boldsymbol{\Sigma}_{XY}}{\sigma_Y\sqrt{\mathbf{a}^\top\boldsymbol{\Sigma}_{XX}\mathbf{a}}}$. With $\mathbf{b} = \boldsymbol{\Sigma}_{XX}^{1/2}\mathbf{a}$, Cauchy–Schwarz gives

$$\mathbf{a}^\top\boldsymbol{\Sigma}_{XY} = \mathbf{b}^\top\boldsymbol{\Sigma}_{XX}^{-1/2}\boldsymbol{\Sigma}_{XY} \le \sqrt{\mathbf{a}^\top\boldsymbol{\Sigma}_{XX}\mathbf{a}}\,\sqrt{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}},$$

so Corr $\le \sqrt{\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}}/\sigma_Y$, with equality iff $\mathbf{a} \propto \boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ (the BLP direction). Then $\text{Cov}(Y, \hat{Y}) = \text{Var}(\hat{Y}) = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\Sigma}_{XY}$ and $\text{Corr}(Y, \hat{Y}) = \sqrt{\text{Var}\hat{Y}}/\sigma_Y$. ■

For two vectors, $\rho(\mathbf{X}_1, \mathbf{X}_2) := \max \text{Corr}(\mathbf{a}^\top\mathbf{X}_1, \mathbf{b}^\top\mathbf{X}_2)$ is the largest canonical correlation (Ch. 8).

4.3 Partial covariance and partial correlation

Partition $\mathbf{X} = (X, \mathbf{Z}^\top)^\top$ with X the predictor of interest and $\mathbf{Z} \in \mathbb{R}^{q-1}$ the others ($\text{Var}\mathbf{Z} \succ 0$), and let $\tilde{X} = X - \text{BLP}(X \mid \mathbf{Z})$, $\tilde{Y} = Y - \text{BLP}(Y \mid \mathbf{Z})$.

Definition 3.17: partial covariance and partial correlation

$$\sigma_{YX\cdot Z} := \text{Cov}(\tilde{Y}, \tilde{X}) = \text{Cov}(Y, X) - \text{Cov}(Y, \mathbf{Z})\text{Var}(\mathbf{Z})^{-1}\text{Cov}(\mathbf{Z}, X),$$

$$\rho_{YX\cdot Z} := \frac{\sigma_{YX\cdot Z}}{\sqrt{\text{Var}(\tilde{Y})\text{Var}(\tilde{X})}} = \text{Corr}(\tilde{Y}, \tilde{X}).$$

In block notation: $\text{pcorr}(\mathbf{X}_1, \mathbf{X}_2; \mathbf{X}_3) = \text{Corr}(\mathbf{X}_1 - \text{BLP}(\mathbf{X}_1 \mid \mathbf{X}_3), \mathbf{X}_2 - \text{BLP}(\mathbf{X}_2 \mid \mathbf{X}_3)) = \text{Corr}(\mathbf{X}_1^*, \mathbf{X}_2^*)$ and the partial covariance matrix $\text{pVar}(\mathbf{X}_1; \mathbf{X}_2) = \text{Var}(\mathbf{X}_1 - \text{BLP}(\mathbf{X}_1 \mid \mathbf{X}_2)) = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$.

Partial covariance formula [Supplement]. $\tilde{Y} = Y - \mu_Y - \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}(\mathbf{Z} - \boldsymbol{\mu}_Z)$ and similarly for $\tilde{X}$. Expanding, $\text{Cov}(\tilde{Y}, \tilde{X}) = \sigma_{YX} - 2\boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX} + \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX} = \sigma_{YX} - \boldsymbol{\Sigma}_{YZ}\boldsymbol{\Sigma}_{ZZ}^{-1}\boldsymbol{\Sigma}_{ZX}$. ■

[Supplement] So the partial covariance matrix of (Y, X) is a Schur complement, and under joint normality it equals the conditional covariance given $\mathbf{Z}$.

Theorem 3.18: Frisch–Waugh–Lovell theorem

Let $\hat{\beta}_X$ be the coefficient of X in the OLS regression of Y on $(X, \mathbf{Z})$, and let $\tilde{y}_i, \tilde{x}_i$ be the residuals from the OLS regressions of Y and X on $\mathbf{Z}$. Then

$$\hat{\beta}_X = \frac{\sum_i \tilde{x}_i\tilde{y}_i}{\sum_i \tilde{x}_i^2},$$

the slope of the residual-on-residual regression (all regressions with intercept; full-column-rank design).

Proof [Supplement]. Let $\mathbf{M}_Z = \mathbf{I} - \mathbf{P}_{[\mathbf{1},\mathbf{Z}]}$. The normal equations give $\mathbf{y} = \hat{\beta}_X \mathbf{x} + [\mathbf{1}, \mathbf{Z}]\hat{\boldsymbol{\gamma}} + \mathbf{e}$ with $\mathbf{e} \perp \mathbf{x}, \mathbf{1}, \mathbf{Z}$. Multiplying by $\mathbf{M}_Z$ ($\mathbf{M}_Z[\mathbf{1}, \mathbf{Z}] = \mathbf{0}$, $\mathbf{M}_Z\mathbf{e} = \mathbf{e}$) gives $\tilde{\mathbf{y}} = \hat{\beta}_X \tilde{\mathbf{x}} + \mathbf{e}$. Multiply by $\tilde{\mathbf{x}}^\top$: $\tilde{\mathbf{x}}^\top \mathbf{e} = \mathbf{x}^\top \mathbf{M}_Z \mathbf{e} = \mathbf{x}^\top \mathbf{e} = 0$, so $\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} = \hat{\beta}_X \tilde{\mathbf{x}}^\top \tilde{\mathbf{x}}$. ■

Example 3.19: mtcars — marginal versus partial correlation

Y = mpg, X = displacement, $\mathbf{Z}$ = (horsepower, weight). The marginal correlation is $\hat{\rho}_{YX} \approx -0.805$, but after adjusting for horsepower and weight $\hat{\rho}_{YX \cdot Z} \approx -0.044$: little linear association remains. The displacement–mpg relation runs mostly through horsepower and weight.

5 Quadratic Forms in Normal Random Vectors

Motivation. Test statistics are typically quadratic forms such as $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu})$. Their χ^2 distributions are the basis of all tests in later chapters.

For $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and symmetric $\mathbf{A}$, a quadratic form is $Y = \mathbf{X}^\top \mathbf{A} \mathbf{X} = \sum_{j,k} X_j a_{jk} X_k$. We want its distribution and conditions for independence. Special case: $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, $\mathbf{A} = \mathbf{I}_p$ gives $Y = \sum X_i^2 \sim \chi^2(p)$.

Remark 3.20: matrix algebra recall

$\mathbf{A}$ is idempotent if $\mathbf{A}^2 = \mathbf{A}$. If $\mathbf{A}$ is symmetric, $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$; if it is also idempotent, its eigenvalues are 0 or 1 and $\operatorname{rank}(\mathbf{A}) = \#\{\text{nonzero eigenvalues}\} = \operatorname{tr}(\mathbf{A})$.

Theorem 3.21: chi-square quadratic forms, $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ (Result 4.7, Case 001)

Let $\mathbf{X} \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I})$ and $\mathbf{A}$ symmetric. Then $Y = \dfrac{\mathbf{X}^\top \mathbf{A} \mathbf{X}}{\sigma^2} \sim \chi^2(m)$ iff $\mathbf{A}$ is idempotent of rank $m \le p$.

Proof. ($\Leftarrow$) $\mathbf{A} = \boldsymbol{\Gamma}\boldsymbol{\Lambda}\boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda} = \operatorname{diag}(1, \ldots, 1, 0, \ldots, 0)$ (m ones). Let $\mathbf{z} = \frac{1}{\sigma}\boldsymbol{\Gamma}^\top \mathbf{X} \sim N(\mathbf{0}, \boldsymbol{\Gamma}^\top \boldsymbol{\Gamma}) = N(\mathbf{0}, \mathbf{I}_p)$; then $Y = \mathbf{z}^\top \boldsymbol{\Lambda} \mathbf{z} = z_1^2 + \cdots + z_m^2 \sim \chi^2(m)$.

($\Rightarrow$) [Supplement] In general $Y = \sum_i \lambda_i z_i^2$ has MGF $\prod_i (1 - 2t\lambda_i)^{-1/2}$; matching $(1 - 2t)^{-m/2}$ near $t = 0$ forces the nonzero λ_i to be m ones. ■

Simple example: $\mathbf{A} = \operatorname{diag}(1, 0, \ldots, 0)$ gives $\mathbf{X}^\top \mathbf{A} \mathbf{X}/\sigma^2 = X_1^2/\sigma^2 \sim \chi_1^2$.

Theorem 3.22: chi-square quadratic forms, general $\boldsymbol{\Sigma}$ (Result 4.7, Case 002)

Let $\mathbf{X} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$ and $\mathbf{A}$ symmetric. Then $Y = \mathbf{X}^\top \mathbf{A} \mathbf{X} \sim \chi^2(m)$ iff either (1) $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent of rank m, or (2) $\boldsymbol{\Sigma}\mathbf{A}$ is idempotent of rank m.

Proof. By the representation theorem $\mathbf{X} = \boldsymbol{\Sigma}^{1/2} \mathbf{z}$, $\mathbf{z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$, so $Y = \mathbf{z}^\top \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2} \mathbf{z}$. By Theorem 3.21, $Y \sim \chi_m^2$ iff $\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$ is (symmetric) idempotent of rank m. Now

$$(\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2})^2 = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2} \iff \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2} \iff \mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A},$$

(multiply by $\boldsymbol{\Sigma}^{-1/2}$ on both sides), and $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{A} \iff (\mathbf{A}\boldsymbol{\Sigma})^2 = \mathbf{A}\boldsymbol{\Sigma}$ (multiply on the right by $\boldsymbol{\Sigma}$ or $\boldsymbol{\Sigma}^{-1}$). Ranks: $\operatorname{rank}(\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}) = \operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}\boldsymbol{\Sigma})$. Since $\boldsymbol{\Sigma}\mathbf{A} = (\mathbf{A}\boldsymbol{\Sigma})^\top$, the same holds for $\boldsymbol{\Sigma}\mathbf{A}$. ■

Example 3.23: distribution of the Mahalanobis distance (Example 4)

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$, then $(\mathbf{X}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X}-\boldsymbol{\mu}) \sim \chi^2(p)$ ($\mathbf{A} = \boldsymbol{\Sigma}^{-1}$ makes $\mathbf{A}\boldsymbol{\Sigma} = \mathbf{I}_p$ idempotent of rank p; apply Case 002 to $\mathbf{X}-\boldsymbol{\mu} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$). Hence $\{\mathbf{x} : (\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \le \chi^2_p(\alpha)\}$ is an ellipsoid of probability $1-\alpha$.

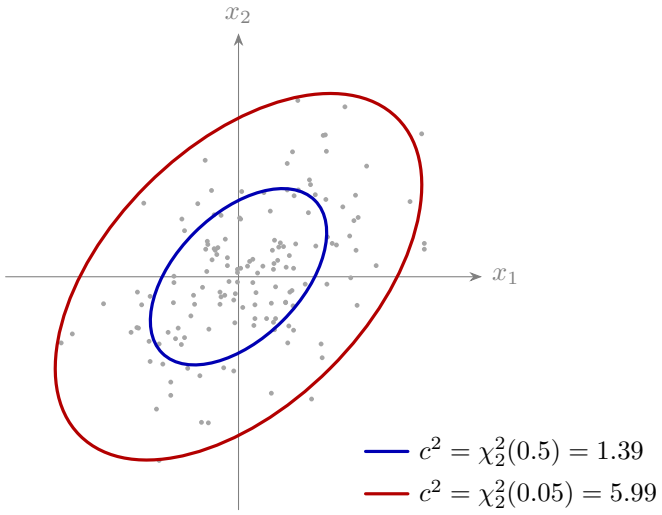


Figure. *Example 4 with* $p = 2$, $\boldsymbol{\mu} = \mathbf{0}$, $\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$*: the ellipses* $(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \le \chi^2_2(\alpha)$ *for* $\alpha = 0.5$ *and* 0.05 *have probability* 0.5 *and* 0.95*. In* 150 *simulated draws,* 54% *and* 96% *fall inside (one draw lies outside the frame).*

Theorem 3.24: independence of linear and quadratic forms (Theorem 9)

Let $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I})$, $\mathbf{A}$ a $p \times p$ symmetric matrix and $\mathbf{B}$ a $k \times p$ matrix. If $\mathbf{BA} = \mathbf{0}$, then $\mathbf{BX}$ and $\mathbf{X}^\top \mathbf{A}\mathbf{X}$ are independent.

Proof. If $\mathbf{A}$ is full rank, $\mathbf{BA} = \mathbf{0}$ forces $\mathbf{B} = \mathbf{0}$: trivial. If $\operatorname{rank}(\mathbf{A}) = m < p$, write $\mathbf{A} = \boldsymbol{\Gamma} \begin{pmatrix} \boldsymbol{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \boldsymbol{\Gamma}^\top$ with $\boldsymbol{\Lambda}_m = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$, all $\lambda_i \neq 0$. From $\mathbf{BA} = \mathbf{0}$, $\mathbf{B}\boldsymbol{\Gamma} \begin{pmatrix} \boldsymbol{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} = \mathbf{0}$. Writing $\mathbf{B}\boldsymbol{\Gamma} = [\mathbf{C}_1, \mathbf{C}_2]$ gives $\mathbf{C}_1 \boldsymbol{\Lambda}_m = \mathbf{0}$, so $\mathbf{C}_1 = \mathbf{0}$ and $\mathbf{B}\boldsymbol{\Gamma} = [\mathbf{0}, \mathbf{C}_2]$. Let $\mathbf{z} = \begin{pmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \end{pmatrix} = \boldsymbol{\Gamma}^\top \mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I})$, so $\mathbf{z}_1 \perp\!\!\!\perp \mathbf{z}_2$. Then

$$\mathbf{BX} = \mathbf{B}\boldsymbol{\Gamma}\mathbf{z} = \mathbf{C}_2 \mathbf{z}_2, \qquad \mathbf{X}^\top \mathbf{A}\mathbf{X} = \mathbf{z}^\top \begin{pmatrix} \boldsymbol{\Lambda}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \mathbf{z} = \mathbf{z}_1^\top \boldsymbol{\Lambda}_m \mathbf{z}_1,$$

functions of $\mathbf{z}_2$ and $\mathbf{z}_1$ respectively, hence independent. ■

Example 3.25: $\bar{X} \perp\!\!\!\perp S^2$

For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ and $\mathbf{X} = (X_1, \ldots, X_n)^\top$, $\bar{X}_n = \frac{1}{n}\mathbf{1}_n^\top \mathbf{X}$ and $S_n^2 = \frac{1}{n-1}\mathbf{X}^\top(\mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top)\mathbf{X}$. With $\mathbf{B} = \frac{1}{n}\mathbf{1}_n^\top$ and $\mathbf{A} = \frac{1}{n-1}\mathbf{H}_n$, $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$, we have $\mathbf{BA} = \mathbf{0}$. With $\mathbf{Z} = (\mathbf{X} - \mu\mathbf{1}_n)/\sigma \sim N_n(\mathbf{0}, \mathbf{I}_n)$ and $\mathbf{H}_n\mathbf{1}_n = \mathbf{0}$, $\bar{X} = \mu + \sigma\mathbf{B}\mathbf{Z}$ and $S^2 = \sigma^2 \mathbf{Z}^\top \mathbf{A}\mathbf{Z}$, so $\bar{X} \perp\!\!\!\perp S^2$ by Theorem 3.24. The multivariate version is in Ch. 4.

Remark 3.26: same argument, two quadratic forms

[Supplement] If $\mathbf{X} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and symmetric $\mathbf{A}, \mathbf{B}$ satisfy $\mathbf{AB} = \mathbf{0}$, then $\mathbf{X}^\top \mathbf{AX} \perp\!\!\!\perp \mathbf{X}^\top \mathbf{BX}$. Indeed $\mathbf{X}^\top \mathbf{BX} = (\mathbf{BX})^\top \mathbf{B}^+ (\mathbf{BX})$ with $\mathbf{B}^+$ the Moore–Penrose inverse ($\mathbf{BB}^+\mathbf{B} = \mathbf{B}$), so it is a function of $\mathbf{BX}$, and $\mathbf{BA} = (\mathbf{AB})^\top = \mathbf{0}$; Theorem 3.24 applies.

6 Selected exercises

Motivation. These exercises sharpen two warnings and add one tool: zero correlation with normal marginals still allows dependence, the normal density factorizes into a marginal and a conditional part, and quadratic forms in normal vectors have explicit means and variances.

Exercise 3.1 (Uncorrelated but dependent; J&W 4.9)

Let $X_1 \sim N(0,1)$, $c > 0$, and $X_2 = -X_1$ if $|X_1| \le c$, $X_2 = X_1$ otherwise.
(a) Show that $X_2 \sim N(0,1)$ for every c.
(b) Show that c can be chosen so that $\mathrm{Cov}(X_1, X_2) = 0$.
(c) For this c, show that X_1 and X_2 are not independent, and conclude that (X_1, X_2) is not bivariate normal.

Solution [Supplement].

(a) For a Borel set B, $P(X_2 \in B) = P(X_1 \in B, |X_1| > c) + P(-X_1 \in B, |X_1| \le c)$. Since $-X_1 \overset{d}{=} X_1$ and $\{|X_1| \le c\} = \{|-X_1| \le c\}$, the second term equals $P(X_1 \in B, |X_1| \le c)$, so $P(X_2 \in B) = P(X_1 \in B)$ (Example 3.4 is the case $c = 1$).

(b)

Step 1. With $g(c) = \mathrm{E}[X_1^2 \mathbf{1}\{|X_1| \le c\}] = \int_{-c}^{c} x^2 \phi(x)\, dx$,

$$\mathrm{Cov}(X_1, X_2) = \mathrm{E}(X_1 X_2) = \mathrm{E}[X_1^2 \mathbf{1}\{|X_1| > c\}] - \mathrm{E}[X_1^2 \mathbf{1}\{|X_1| \le c\}] = 1 - 2g(c).$$

Step 2. g is continuous, $g(c) \to 0$ as $c \downarrow 0$, and $g(c) \to \mathrm{E}X_1^2 = 1$ as $c \to \infty$ (monotone convergence). So $h(c) = 1 - 2g(c)$ is continuous with limits 1 and -1, and by the intermediate value theorem $h(c^*) = 0$ for some $c^* > 0$ (numerically $c^* \approx 1.54$).

(c)

Step 1. $|X_2| = |X_1|$, so $X_2^2 = X_1^2$. If $X_1 \perp\!\!\!\perp X_2$, then $X_1^2 \perp\!\!\!\perp X_2^2$ and $\mathrm{Var}(X_1^2) = \mathrm{Cov}(X_1^2, X_2^2) = 0$, contradicting $\mathrm{Var}(X_1^2) = \mathrm{E}X_1^4 - 1 = 2$.

Step 2. If (X_1, X_2) were bivariate normal, $\mathrm{Cov}(X_1, X_2) = 0$ would imply independence by Theorem 3.10. Hence it is not jointly normal, although both marginals are $N(0,1)$. ■

Exercise 3.2 (Factorizing the normal density; J&W 4.13)

Let $\mathbf{X} = (\mathbf{X}_1^\top, \mathbf{X}_2^\top)^\top \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$ partitioned as in Theorem 3.12, and $\boldsymbol{m}(\mathbf{x}_2) = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2)$.
(a) Show $|\boldsymbol{\Sigma}| = |\boldsymbol{\Sigma}_{22}|\,|\boldsymbol{\Sigma}_{11\cdot 2}|$.
(b) Show

$$(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = \{\mathbf{x}_1 - \boldsymbol{m}(\mathbf{x}_2)\}^\top \boldsymbol{\Sigma}_{11\cdot 2}^{-1} \{\mathbf{x}_1 - \boldsymbol{m}(\mathbf{x}_2)\} + (\mathbf{x}_2 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{x}_2 - \boldsymbol{\mu}_2).$$

(c) Using only the density, deduce the marginal law of $\mathbf{X}_2$ and the conditional law of $\mathbf{X}_1$ given $\mathbf{X}_2 = \mathbf{x}_2$.

Solution [Supplement].

(a) This is the first determinant formula of Theorem 2.6 with $\mathbf{A} = \boldsymbol{\Sigma}$.

(b)

Step 1. Let $\mathbf{E} = \begin{pmatrix} \mathbf{I} & -\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$. Since $\boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^\top$ and $\boldsymbol{\Sigma}_{22}$ is symmetric, the elimination identity in the proof of Theorem 2.6 reads $\mathbf{E}\boldsymbol{\Sigma}\mathbf{E}^\top = \mathbf{D} := \operatorname{diag}(\boldsymbol{\Sigma}_{11\cdot 2}, \boldsymbol{\Sigma}_{22})$.

Step 2. Both $\mathbf{E}$ and $\mathbf{D}$ are invertible, so $\boldsymbol{\Sigma}^{-1} = \mathbf{E}^\top \mathbf{D}^{-1}\mathbf{E}$ and the quadratic form equals $\{\mathbf{E}(\mathbf{x} - \boldsymbol{\mu})\}^\top \mathbf{D}^{-1}\{\mathbf{E}(\mathbf{x} - \boldsymbol{\mu})\}$.

Step 3. $\mathbf{E}(\mathbf{x} - \boldsymbol{\mu}) = (\mathbf{x}_1 - \boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2),\ \mathbf{x}_2 - \boldsymbol{\mu}_2) = (\mathbf{x}_1 - \boldsymbol{m}(\mathbf{x}_2),\ \mathbf{x}_2 - \boldsymbol{\mu}_2)$, and $\mathbf{D}^{-1}$ is block diagonal, which gives the identity.

(c)

Step 1. $\boldsymbol{\Sigma}_{11\cdot 2} = \{(\boldsymbol{\Sigma}^{-1})_{11}\}^{-1}$ is the inverse of a principal block of a positive definite matrix, hence positive definite. With $r = \dim \mathbf{X}_1$, $s = \dim \mathbf{X}_2$, (a) and (b) give

$$f(\mathbf{x}_1, \mathbf{x}_2) = g(\mathbf{x}_1 \mid \mathbf{x}_2)\, h(\mathbf{x}_2),$$

where $g(\cdot \mid \mathbf{x}_2)$ is the $N_r(\boldsymbol{m}(\mathbf{x}_2), \boldsymbol{\Sigma}_{11\cdot 2})$ density and h the $N_s(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$ density (the constants split as $(2\pi)^{-p/2} = (2\pi)^{-r/2}(2\pi)^{-s/2}$).

Step 2. For each $\mathbf{x}_2$, $\int g(\mathbf{x}_1 \mid \mathbf{x}_2)\, d\mathbf{x}_1 = 1$, so the marginal density of $\mathbf{X}_2$ is $\int f\, d\mathbf{x}_1 = h(\mathbf{x}_2)$: $\mathbf{X}_2 \sim N_s(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$.

Step 3. The conditional density is $f(\mathbf{x}_1, \mathbf{x}_2)/h(\mathbf{x}_2) = g(\mathbf{x}_1 \mid \mathbf{x}_2)$, which is Theorem 3.12 again, now obtained from the density. ■

Exercise 3.3 (Mean and variance of a normal quadratic form)

Let $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ ($\boldsymbol{\Sigma} \succeq 0$) and $\mathbf{A}$ symmetric. Show

$$\mathrm{E}(\mathbf{X}^\top \mathbf{A}\mathbf{X}) = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}, \qquad \operatorname{Var}(\mathbf{X}^\top \mathbf{A}\mathbf{X}) = 2\operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}) + 4\boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\mu}.$$

Solution [Supplement].

Step 1 (mean). $\mathrm{E}(\mathbf{X}^\top \mathbf{A}\mathbf{X}) = \mathrm{E}\operatorname{tr}(\mathbf{A}\mathbf{X}\mathbf{X}^\top) = \operatorname{tr}\{\mathbf{A}(\boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top)\} = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}$ (normality is not needed here).

Step 2 (representation). Let $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\boldsymbol{\Sigma}^{1/2}$ the symmetric square root. By Theorem 3.8, $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$ is normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and a normal law is determined by its mean and covariance (Theorem 3.5), so $\mathbf{X} \overset{d}{=} \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$. Let $\mathbf{B} = \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}^{1/2} = \mathbf{P}\boldsymbol{\Lambda}\mathbf{P}^\top$ (spectral decomposition), $\mathbf{W} = \mathbf{P}^\top \mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\mathbf{b} = \mathbf{P}^\top \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\mu}$. Then

$$\mathbf{X}^\top \mathbf{A}\mathbf{X} \overset{d}{=} \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu} + 2\mathbf{b}^\top \mathbf{W} + \mathbf{W}^\top \boldsymbol{\Lambda}\mathbf{W} = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu} + \sum_{i=1}^{p} (2b_i W_i + \lambda_i W_i^2).$$

Step 3 (variance). The summands are independent. With $\mathrm{E}W_i^3 = 0$ and $\operatorname{Var}(W_i^2) = 3 - 1 = 2$,

$$\operatorname{Var}(2b_i W_i + \lambda_i W_i^2) = 4b_i^2 + 2\lambda_i^2 + 4b_i\lambda_i \operatorname{Cov}(W_i, W_i^2) = 4b_i^2 + 2\lambda_i^2.$$

Step 4. $\sum_i \lambda_i^2 = \operatorname{tr}(\mathbf{B}^2) = \operatorname{tr}(\boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}^{1/2}) = \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma})$ and $\sum_i b_i^2 = \mathbf{b}^\top \mathbf{b} = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\Sigma}^{1/2}\mathbf{P}\mathbf{P}^\top \boldsymbol{\Sigma}^{1/2}\mathbf{A}\boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\mu}$, which gives the variance formula.

Check: $\boldsymbol{\mu} = \mathbf{0}$ and $\mathbf{A} = \boldsymbol{\Sigma}^{-1}$ give mean p and variance $2p$, those of χ_p^2 (Example 3.23). ■